4. What is the contribution in Feynman diagrams from the contraction of the derivative $\partial_\mu \psi_\ell(x)$ of a Dirac field with the adjoint $\psi_m^\dagger(y)$ of the field?

5. Use the theorem of Section 6.4 to give expressions for the vacuum expectation values of Heisenberg picture operators $(\Psi_0, \Phi(x)\Psi_0)$ and $(\Psi_0, T\{\Phi(x), \Phi(y)\}\Psi_0)$ in the theory of Problem 1, to orders g and g^2, respectively.

References

1. F. J. Dyson, *Phys. Rev.* **75**, 486, 1736 (1949).

2. The formal statement of this result is known as *Wick's theorem*; see G. C. Wick, *Phys. Rev.* **80**, 268 (1950).

3. I do not know who first proved this theorem. It was known in the early 1950s to several theorists, including M. Gell-Mann and F. E. Low.

4. J. Schwinger, *Phys. Rev.* **82**, 914 (1951).